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On the Handoff Mechanism of IEEE 802.11s Mesh Protocol

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Abstract

IEEE 802.11s provides support for Wi-Fi mesh networks and it has been identified as a potential candidate for use in mobile environments where packets will have to be transmitted over multiple hops of a wireless connection. The ability of the routing layer to respond rapidly to changes in link conditions is key to how well a mesh network operates when there are mobility issues. Rapid and short-duration variations in channel quality during vehicular travel can cause frequent route changes, thereby disrupting ongoing communications and causing long-term instability. The objective of this project was to simulate using NS-3, a simple three-node mesh topology comprised of two roadside mesh nodes (one fixed at each end) and a mobile vehicle-mounted node with speeds of 20 km/hr, 50 km/hr, and 80 km/hr.

Our results show that the default HWMP airtime-based routing metric is highly sensitive to transient link variations. In the baseline configuration, we observed packet loss approaching 80%, highly irregular round-trip times, sharp drops in throughput along transition regions, and significant jitter spikes — all consequences of the metric reacting too aggressively to momentary link changes. To address this, we propose the Hysteresis Airtime Link Metric (H-ALM), an extension of the standard airtime metric that blends the current airtime value with a previously stabilized estimate using a configurable weighting factor α . The H-ALM architecture provides for the H-ALM system to be able to respond to real link degradation as well as filter out transient link conditions which would otherwise force the H-ALM system to unnecessarily switch routes back-and-forth (ping-pong). H-ALM shows significant and measurable performance gains: average packet loss is significantly reduced by about 91% with the RTT being much more stable; the H-ALM system also shows steady increases in both TCP and UDP throughputs, and jitter has been shown to be under 280 ms. Thus, these test results show that an appropriate use of a very light weight hysteresis function can significantly reduce handoff delay in IEEE 802.11s-based vehicle mesh networks.

Keywords: Wireless mesh networks, IEEE 802.11s, mobility, hard handoff, airtime link metric, ping-pong effect, hysteresis, NS-3.

1 Introduction

Conventional IEEE 802.11 deployments can expand coverage by adding more access points, yet wider coverage alone does not guarantee smooth communication for a moving user. Once a session spans multiple coverage regions, the network must continue forwarding traffic while keeping interruption time, delay, and loss within acceptable limits [1]. This requirement becomes especially important for delay-sensitive applications such as voice, interactive video, and real-time data exchange.

IEEE 802.11s is relevant in this context because it extends Wi-Fi with mesh support and multi-hop forwarding. Instead of relying only on a direct association with a single access point,

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traffic may traverse intermediate mesh routers before reaching its destination. The appeal of that architecture for roadside networks, campus-wide wireless deployments, public transportation systems, as well as vehicle-to-vehicle (and/or vehicle-to-infrastructure) communications lies in its ease-of-deployment; however, this same architecture makes mobility management more directly connected to routing decisions than would be found in a standard single-hop WLAN.

A problem arises when the movement of an object causes frequent changes in how good a particular link appears to be. When an object passes from area of a network's coverage into an adjacent area of coverage, there will be many temporary variations in the quality of the link due to signal strength issues, radio frequency interference, or varying data rates. If the routing function responds to all these minor variations, then the mobile device will frequently switch back-and-forth among alternative paths. In practice, this produces hard-handoff instability and the familiar ping-pong effect, which is often accompanied by higher packet loss, larger RTT variation, and degraded throughput [2, 3, 4].

In IEEE 802.11s, route selection is governed by the Hybrid Wireless Mesh Protocol (HWMP), whose default path choice is driven by the Airtime Link Metric (ALM) [5, 6, 7]. ALM is designed to favor paths that consume less channel time, but under mobility that same sensitivity can become a weakness. A transient improvement in a link may be enough to trigger a switch even when the change is not persistent. Several earlier studies [2, 8, 9, 10] report behavior consistent with this concern, and our own NS-3 observations show the same pattern: without a stabilizing mechanism, the mobile node can switch routes more often than necessary.

In this research paper, we investigate this issue in an environment of vehicles; we also propose a very simple extension to the typical measurement metrics.

We present a Hysteresis based Air Time Link Metric (H-ALM), which extends the current air time measurement for links with a history part. We are not looking to create a completely unresponsive protocol, rather to reduce the amount of route update requests made as a result of temporary changes that do not indicate a permanent alteration to the quality of the route. We use simulations of a three-node IEEE 802.11s topology within the NS-3 simulator to test our proposal. The nodes include two stationary roadside mesh routers and one node mounted on a vehicle which moves throughout its overlap of service area with the stationary nodes [11]. Performance of handover is evaluated by packet loss, RTT, jitter, and TCP/UDP throughput at speeds between 20 km/h to 80 km/h. Experimental results show that mobility enhances route oscillation in baseline HWMP and that even a small amount of hysteresis can improve stability significantly.

The main contributions of this paper are as follows:

1. We propose a hysteresis-aware extension of the airtime link metric for HWMP using a tunable factor α to combine current and historical path information.
2. We conducted a comparative study on IEEE 802.11s handoff behavior under vehicle movement with speed ranges from 20 km/h to 80 km/h by means of packet loss, RTT, throughput, and jitter as evaluation metrics using NS-3.
3. Discussion on what the results imply for practical mobility-aware mesh routing, including responsiveness vs. Route stability trade-off.

The rest of the paper is organized as follows. Section 2 summarizes related work. Section 3 introduces proposed method, system model and simulation set up. Section 4 reports and interprets experimental results. Section 5 closes the paper and outlines possible future extensions.

2 Related Work

Research on wireless handoff has traditionally focused on infrastructure Wi-Fi, where the main goal is to move a client from one access point to another with as little delay as possible [2, 12].

In general, common goals for improving roaming in mobile environments focus on shortening authentication times and reducing channel-scan delays and re-associating [3, 13, 14, 15, 16, 17]. As an example, techniques like optimized (or "fast") scanning and unified addressing or "fast" re-associate protocols help improve roaming performance [18, 19, 20, 21, 22]. However, these solutions are generally designed with single-hop wireless LANs in mind. Other researchers studying vehicular and large-scale area roaming (for example, using IEEE 802.11p, LTE systems, and/or other types of managed-wireless infrastructures) have developed many of the same kinds of ideas using threshold based handoffs and decision-making algorithms based upon prediction and/or explicit network-wide coordination [23, 24].

The challenge in IEEE 802.11s is different because mobility affects not only association but also forwarding paths. Mesh network links have characteristics that make the effect of an ill-timed route update potentially much worse than with traditional point-to-point wireless networks. For instance, if there is a simultaneous degradation of link quality while one node determines it would be better off to use another node, its bad decision will not be limited to that particular link; rather, it could affect multiple hops. While previous research has identified delay as a significant consideration in performing 802.11 handoffs [25], mesh networks raise additional issues relative to maintaining consistent paths among multiple hops, potential queueing delays due to contention and handoffs occurring at intermediate nodes, and the possibility of handoffs causing routes to "oscillate" (i.e., alternate) at various intermediate nodes.

More recently, several new solutions demonstrate the ability of highly coordinated Wi-Fi control systems to produce very low latency when performing handoffs [26]. However, these are not applicable to the standards-based IEEE 802.11s implementation of HWMP since this system is designed to operate in a distributed fashion and therefore does not provide for centralized control of all routing decisions.

In the same body of research as Mesh-Routing, Gokalgandhi et al. introduced the RLLR metric and demonstrated a link to improved system stability using SNR and MCS-aware routing metrics for wireless networks. The relevance of this finding is based on its validation that selecting stable paths over time in response to changing wireless conditions is an important factor in determining overall performance. Other researchers in the domain of vehicular Wi-Fi are also improving reliability and performance with the use of mechanisms such as "make before break" protocols, dual radio configurations, and other similar hardware-assisted solutions to lower both packet loss and delay [27, 28]. These approaches can work well, but they often assume extra hardware capability or additional complexity.

Related ideas also appear in ICN-based mobility handling and heterogeneous handoff algorithms, where predictive thresholds, dwell time, and context-aware control are used to suppress unnecessary switching [29, 30, 31]. Likewise, RSSI-driven and Wi-Fi 6-oriented methods [32, 33, 34, 35] aim to improve access-point choice and coordination. However, relatively few of these studies focus on the specific question addressed here: how the airtime-based decision logic of HWMP behaves when a mobile node crosses overlapping mesh coverage regions.

The literature therefore leaves a useful gap. Existing work explains how to reduce handoff delay, improve link selection, or coordinate network-side control, but it does not fully examine mobility-induced route oscillation in IEEE 802.11s itself. The present paper addresses that gap by studying the behavior of HWMP under vehicular movement and by introducing a lightweight hysteresis term intended to stabilize route selection.

3 Proposed H-ALM Framework and System Model

We target the instability that appears when HWMP reacts too quickly to short-lived link changes. The central idea is straightforward: retain the standard airtime-based routing structure, but modify the metric so that each new decision is influenced not only by the current observation but also by the recent history of that path. This gives the route-selection process a

small amount of memory and helps prevent repeated back-and-forth switching during mobility.

3.1 Standard Airtime Link Metric (ALM)

In standard HWMP, route discovery and link-cost evaluation are performed together. The source broadcasts a Path Request (PREQ), and each forwarding mesh node adds its local cost contribution to the value already carried by the request. The cumulative path cost is therefore updated as follows:

$$M_{cum} = M_{local} + M_{received} \quad (3.1)$$

After the PREQ reaches the destination, a Path Reply (PREP) returns with the accumulated metric for the discovered path. The source finally chooses the route with the smallest overall airtime cost.

The airtime cost defined in IEEE 802.11s Draft D3.0 [36, 37] is given by

$$C_a = \frac{\left(O + \frac{B_t}{r}\right) \left(\frac{1}{1-\text{FER}}\right)}{0.01} \quad (3.2)$$

In Eq. 3.2, O denotes the fixed channel-access overhead introduced by PHY/MAC operation and contention, expressed in μs . The quantity B_t is the 8192-bit test frame defined for the metric, and r is the link rate in Mbps. The multiplicative term $\frac{1}{1-\text{FER}}$ reflects the extra transmission effort implied by the frame error rate. The denominator 0.01 is the normalization term used in the draft standard.

3.2 The Proposed Hysteresis Airtime Link Metric (H-ALM)

To make route changes less sensitive to brief channel fluctuations, we define a hysteresis-enhanced airtime metric:

$$C_h = \alpha \cdot C_a + (1 - \alpha) \cdot C_{past} \quad (3.3)$$

Here, C_a is the instantaneous airtime value obtained from Eq. 3.2, while C_{past} stores the previously stabilized value of the same metric. The weighting factor $\alpha \in (0, 1)$ determines how much confidence is given to the newest observation.

The interpretation of α is intuitive. Values close to 1 make the metric behave almost like the default airtime cost and therefore preserve fast reaction to link changes. Values close to 0 put more emphasis on history and make the route update process more conservative. The useful operating region lies between these extremes, where the protocol can still adapt to genuine degradation but is less likely to switch because of a short-lived fluctuation.

3.3 Algorithm Implementation

Algorithm 1 summarizes the metric update procedure used in our implementation.

Algorithm 1 Hysteresis Airtime Link Metric (H-ALM) Calculator

Require: Channel-access overhead O , test-frame size B_t , current data rate r , frame error rate FER, hysteresis factor $\alpha \in (0, 1)$, previous stabilized metric C_{past} , maximum metric value C_{max} , initialization flag $first_time$

Ensure: Hysteresis-adjusted airtime metric C_h

```

1: if FER = 1 then
2:   return  $C_{max}$                                      ▷ Link is unusable
3: end if
4:  $C_a \leftarrow \frac{(O + \frac{B_t}{r}) \left( \frac{1}{1-FER} \right)}{0.01}$ 
5: if  $first\_time$  then
6:    $C_h \leftarrow C_a$ 
7:    $first\_time \leftarrow \text{false}$ 
8: else
9:    $C_h \leftarrow \alpha \cdot C_a + (1 - \alpha) \cdot C_{past}$ 
10: end if
11:  $C_{past} \leftarrow C_h$ 
12: return  $C_h$ 

```

3.4 Simulation Setup and System Layout

We evaluate the proposed method in NS-3 using a compact IEEE 802.11s topology designed to expose handoff behavior clearly. The network contains three mesh nodes: two fixed roadside routers, labeled Node 1 and Node 2, and one mobile node attached to a vehicle. The fixed nodes are placed at an approximate height of 11 feet and use 9 dBi omnidirectional antennas so that their coverage regions overlap. As the vehicle moves through that shared region, the mobile node encounters the exact conditions under which route switching and ping-pong behavior are most likely to occur. Figure 1 illustrates the layout.

3.4.1 Testbed Configuration

Table 1 lists the main simulation parameters used in the testbed.

Table 1: Testbed Configuration Parameters

Parameter	Value
Experimental field length	1084 m
MAC layer protocol	IEEE 802.11s
Physical layer configuration	IEEE 802.11-based mesh PHY (scenario-dependent)
Antenna characteristics	9 dBi omnidirectional
Communication modes	Direct & multihop
Vehicle speeds	20, 50, 80 km/h
Operating band/channel	Configured according to the evaluated PHY scenario
Traffic types	ICMP (ping), iperf (UDP, TCP)
Transmit power	500 mW
Packet interval time	1 ms

3.4.2 Propagation Models and Assumptions

The simulation adopts the following modeling assumptions:

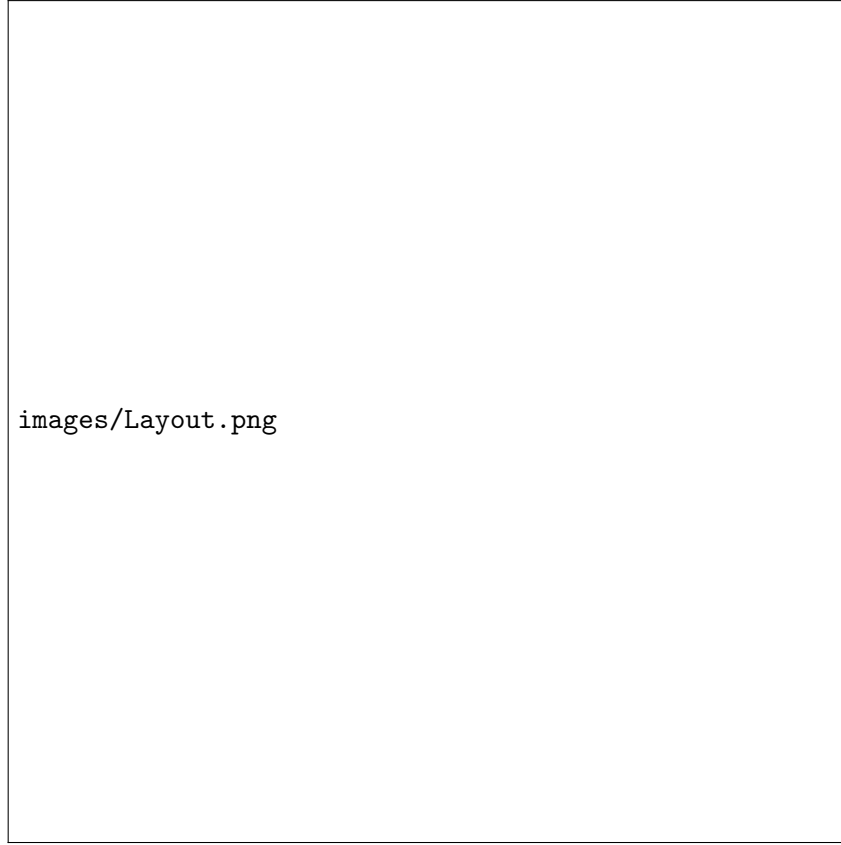


Figure 1: Experimental Field Layout

• **Propagation Loss Models:**

- log-distance path loss model: This model serves as a baseline representation for generic mesh communication conditions:

$$PL_{LD}(d) = PL(d_0) + 10n \log_{10} \left(\frac{d}{d_0} \right) + X_{\sigma} \quad (3.4)$$

where $d_0 = 1$ m, $n = 3.0$ is the path loss exponent, and $X_{\sigma} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 4$ dB.

- ThreeGppV2vHighwayPropagationLossModel for highway and urban scenarios:

$$PL_{3GPP}(d) = 40 \log_{10}(d) + 13.47 - 14 \log_{10}(h_{Tx}) - 14 \log_{10}(h_{Rx}) \quad (3.5)$$

where the antenna heights are fixed at $h_{Tx} = h_{Rx} = 1.5$ m.

- **Node Configuration:** All stationary nodes use the default receiver sensitivity of -101 dBm. In addition, all nodes operate with the IEEE 802.11s HWMP protocol, and the initial hysteresis parameter is statically set to $\alpha = 0.5$. This parameter is configurable and may be adjusted to match different network requirements.

3.5 Key Performance Metrics to Evaluate Handoff

We evaluate handoff quality in IEEE 802.11s mesh networks using the following performance measures:

- **Packet Loss**

Packet loss measures delivery reliability by calculating the proportion of transmitted packets that do not reach the destination:

$$\text{Packet Loss (\%)} = \left(1 - \frac{N_{\text{recv}}}{N_{\text{send}}}\right) \times 100\% \quad (3.6)$$

where N_{send} and N_{recv} are the total numbers of transmitted and successfully received ICMP packets. Higher values indicate poorer delivery performance.

- **Round-Trip Time (RTT)**

RTT is used as a delay indicator and is smoothed through the following simple moving-average formulation:

$$\overline{\text{RTT}}_n = \begin{cases} \text{RTT}_1 & n = 1 \\ \frac{\overline{\text{RTT}}_{n-1} + \text{RTT}_n}{2} & n > 1 \end{cases} \quad (3.7)$$

This formulation reduces the effect of short-lived delay spikes during mobility events. Although it is simpler than the TCP estimator defined in RFC 6298 [38], it is sufficient for comparing latency trends across the evaluated scenarios.

- **Jitter**

Jitter captures packet-delay variation and is particularly relevant for real-time traffic:

$$J_n = |D_n - D_{n-1}|, \quad \bar{J}_n = \frac{\bar{J}_{n-1} + J_n}{2} \quad (3.8)$$

where D_n denotes the one-way delay of the n^{th} packet.

- **Throughput**

Throughput represents the effective rate of successful data delivery:

$$\text{Throughput (Mbps)} = \frac{8 \times \Delta B}{\Delta t \times 10^6} \quad (3.9)$$

where ΔB is the number of bytes delivered during the interval Δt .

- **Received Signal Strength (RSSI)**

RSSI is used as an indicator of received signal quality and is tracked here through an averaged expression:

$$\text{RSSI (dBm)} = 10 \log_{10} \left(\frac{P_{\text{recv}}}{1 \text{ mW}} \right), \quad \overline{\text{RSSI}}_n = \frac{\overline{\text{RSSI}}_{n-1} + \text{RSSI}_n}{2} \quad (3.10)$$

- **Airtime Link Metric (ALM)**

ALM is the IEEE 802.11s routing metric defined in Eq. 3.2.

4 Simulation Results and Discussion

This section compares three routing configurations—baseline HWMP, ALM+HWMP, and H-ALM+HWMP—at vehicle speeds of 20, 50, and 80 km/h. In each subplot, the horizontal axis gives the distance traveled by the vehicle and the vertical axis gives the corresponding performance value. The goal is to show how mobility affects route stability and how that instability appears in RSSI, packet loss, RTT, throughput, and jitter.

4.1 Received Signal Strength Indicator (RSSI)

Figure 2 shows the RSSI observed by the mobile node relative to Node 1 and Node 2. The overall pattern follows the expected geometry of the scenario: the signal from one roadside node decreases as the vehicle moves away, while the signal from the other increases as the vehicle approaches it. The three routing variants produce almost the same RSSI traces, which is reasonable because RSSI is determined mainly by propagation and node position rather than by routing logic. Across all tested speeds, distance remains the dominant factor and speed contributes only limited short-term fluctuation.



Figure 2: RSSI versus distance

4.2 Packet Loss Analysis

As speed increases, channel conditions change more quickly and unstable handoff becomes more likely. Figure 3 compares packet-loss behavior for the three routing variants.

Under baseline HWMP (Figure 3, first row), packet loss rises sharply once the mobile node enters the unstable transition region, especially at longer distances. This pattern is consistent with repeated switching between Node 1 and Node 2. Standard ALM+HWMP (second row) improves the average behavior, but the loss curves still remain irregular and do not fully remove instability.

The clearest improvement appears with H-ALM+HWMP (third row). By damping route changes that are triggered only by short-lived fluctuations, the proposed metric preserves delivery over a much larger portion of the route. Table 2 shows that the average packet-loss reduction relative to baseline HWMP is about 91% across the evaluated speeds. The key gain is therefore not only lower average loss, but a far more stable delivery pattern in the handoff region.



Figure 3: Packet loss

The reported 91% reduction in packet loss represents the average improvement achieved by H-ALM+HWMP over baseline HWMP across all evaluated speeds and distances up to 600 m.

Table 2: Average Packet Loss Across All Distances (100-600 m)

Speed	HWMP Loss	ALM+HWMP Loss	H-ALM Loss	Reduction
20 km/h	81.5%	31.5%	10.0%	90.53%
50 km/h	83.0%	32.5%	8.5%	92.23%
80 km/h	78.0%	29.2%	10.3%	90.23%
Grand Average	80.8%	31.1%	9.6%	91%

4.3 Round Trip Time (RTT)

In this topology, RTT is influenced more by route instability than by RSSI alone. Baseline HWMP produces the least stable delay behavior. In Figure 4 (first row), large RTT excursions begin to appear once the vehicle reaches the overlap region where either roadside node may temporarily look preferable. The same pattern remains visible at 50 and 80 km/h, showing that default HWMP is vulnerable to mobility-induced oscillation.

Some of the most extreme values in the raw traces should be interpreted as simulation-side effects of prolonged queuing, failed route discovery, or transient routing loops rather than as realistic application-level delay. In a real deployment, many of those packets would be dropped instead of being delivered after very long waits. Standard ALM+HWMP (second row) improves a few portions of the route, but large latency spikes still remain, so the instability problem is not fully resolved.

H-ALM+HWMP (third row) produces a substantially cleaner RTT profile. At 20 km/h, practical RTT stays near the lower range for a longer part of the route, and the largest usable-delay peaks are much smaller. The same stabilizing effect appears at 50 and 80 km/h. Table 3 also shows that RTT has only a weak correlation with RSSI, which supports the interpretation that delay variation is dominated by route-selection behavior rather than by signal strength alone.

Under stable communication, RTT remains close to 1 ms for all schemes. The major delay inflation appears during handoff, where repeated switching can keep RTT elevated for seconds at a time in HWMP and, to a lesser degree, in ALM+HWMP. By adding hysteresis, H-ALM suppresses many of those repeated transitions and keeps RTT within a more practical range over a wider operating distance. Table 4 highlights the same point at 80 km/h: H-ALM extends the stable range, lowers handoff frequency, and shortens RTT-spike duration.

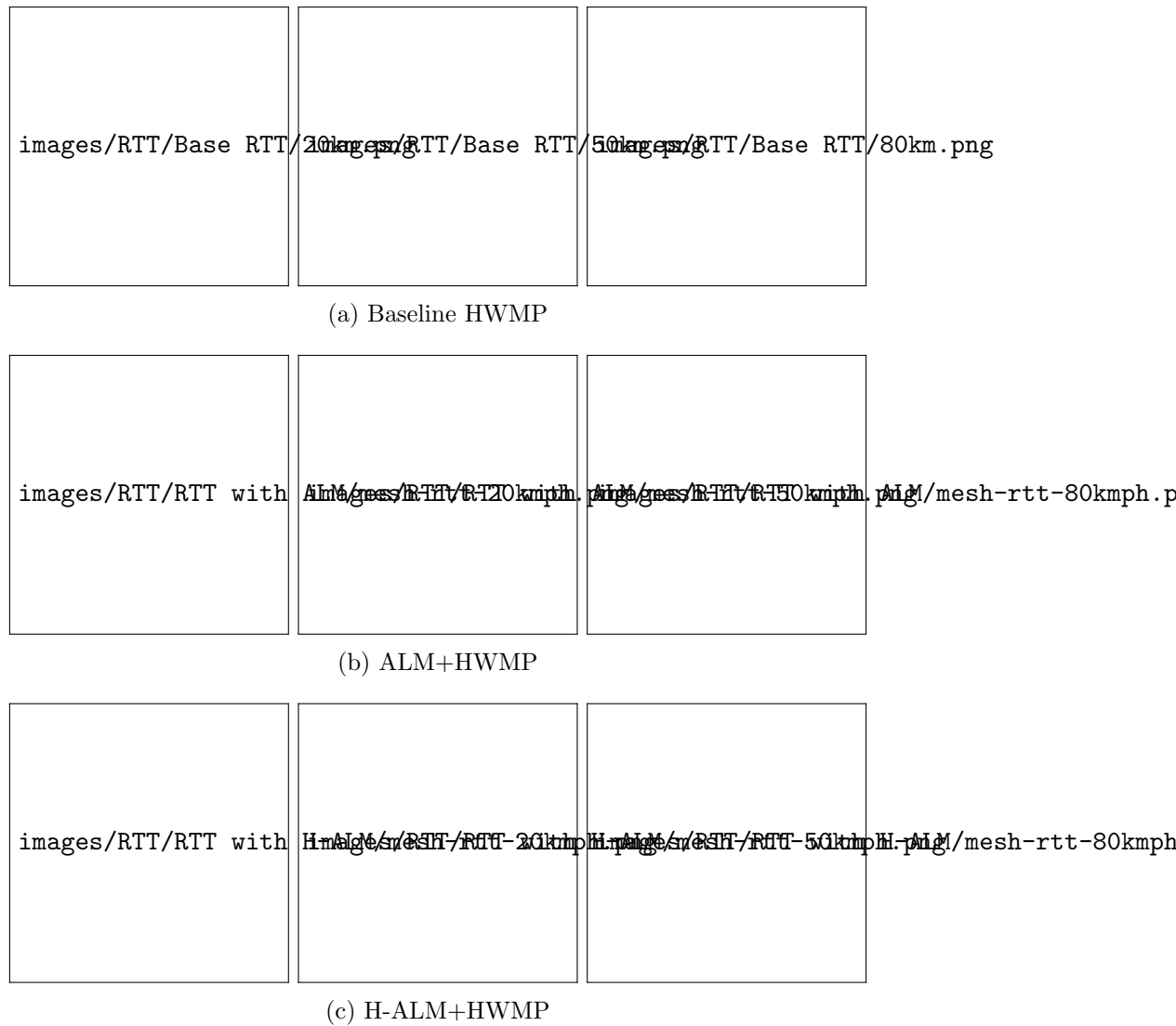


Figure 4: Round Trip Time (RTT)

Table 3: RTT-RSSI Correlation Analysis

Protocol	RTT-RSSI Pearson r	p-value
HWMP	0.28	< 0.05
ALM+HWMP	0.19	< 0.05
H-ALM+HWMP	0.11	< 0.10

Table 4: RTT and Handoff Performance Comparison at 80 km/h

Metric	HWMP	ALM+HWMP	H-ALM+HWMP
Peak practical RTT @ 80 km/h	> 1000 ms	≈ 1500 ms	≤ 1000 ms
Max stable range	< 300 m	< 400 m	> 500 m
Handoffs/km	14.7	9.2	2.1
RTT spike duration	5–10 s	2–5 s	< 0.5 s

4.4 Throughput (TCP and UDP) Evaluation

Figures 5 and 6 compare TCP and UDP throughput for the three routing variants at 20, 50, and 80 km/h.

Baseline HWMP produces the least stable throughput. Because route changes occur frequently in the transition region, TCP has difficulty maintaining congestion-window growth and UDP experiences abrupt delivery drops during ping-pong switching. The problem becomes more visible as distance increases and is especially clear near the far end of the route.

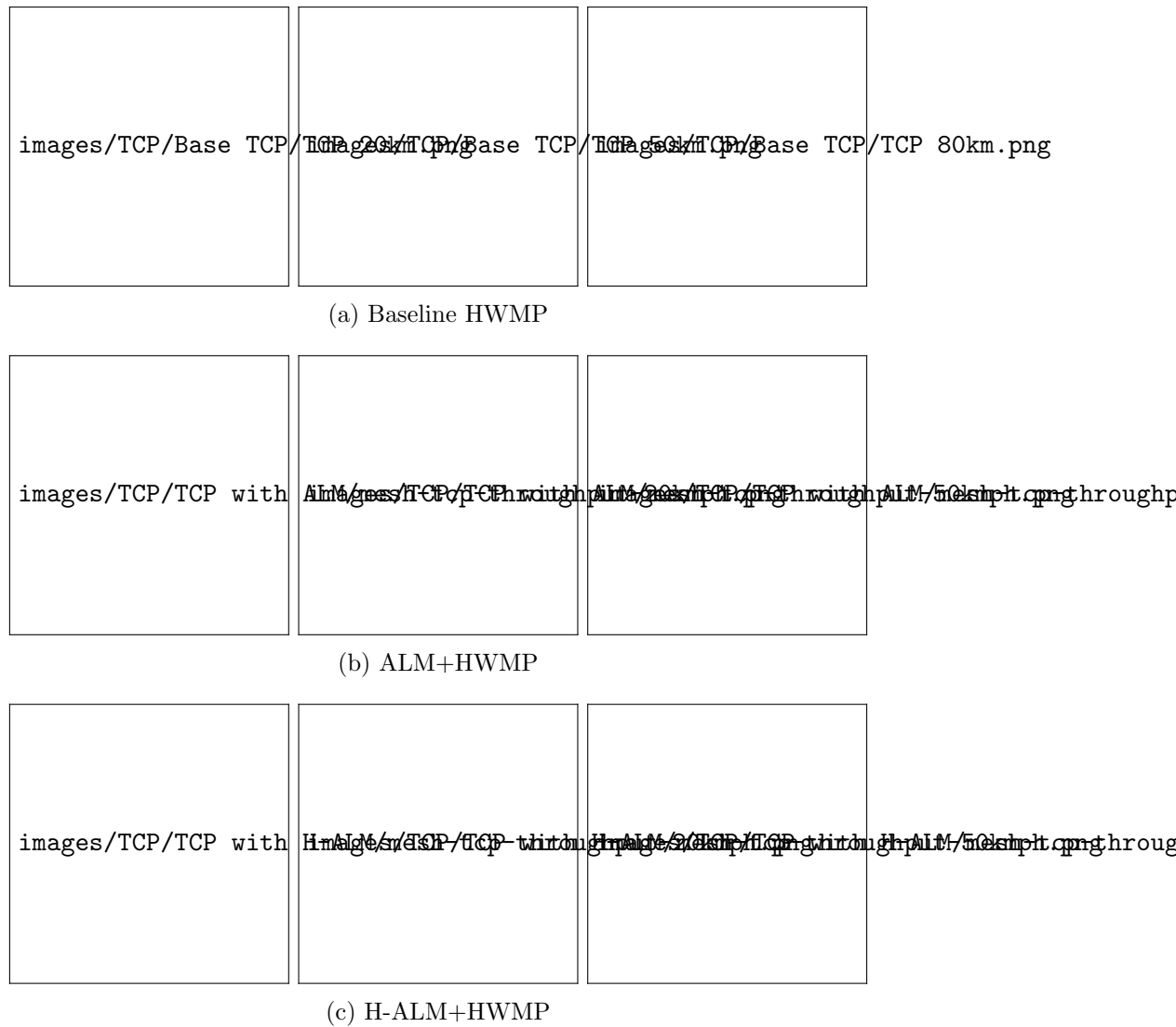
Adding standard ALM improves path choice to some extent because lower-airtime links are selected more aggressively. This reduces a number of severe collapses and yields moderate gains for both TCP and UDP. Even so, the throughput traces remain irregular, particularly at higher speeds, which indicates that instantaneous airtime preference alone is still too reactive.

The H-ALM variant provides the most consistent throughput behavior. With $\alpha = 0.5$, the routing process no longer responds as strongly to brief fluctuations, so transport sessions are interrupted less often. TCP retains higher usable throughput over longer distances, and UDP delivery becomes more continuous across the same regions. The benefit is visible not only in larger values, but also in smoother traces.

Cross-layer perspective: the throughput results reinforce the routing interpretation above. Under baseline HWMP, route oscillation propagates upward and directly disturbs transport behavior. TCP reacts through congestion-control dynamics, while UDP continues sending and therefore exposes delivery instability more openly. ALM reduces part of this effect but does not remove it. H-ALM lowers handoff frequency by roughly 80–85%, enabling both transport protocols to operate over a more stable path. As shown in Table ??, that translates into better TCP/UDP coexistence and a smaller multi-hop throughput penalty.

Table 5: Cross-Layer Performance Comparison of HWMP, ALM+HWMP, and H-ALM+HWMP

Issue	HWMP	ALM+HWMP	H-ALM+HWMP
TCP/UDP Synchronization	TCP masks UDP instability (TCP: 5 Mbps, UDP drops to 2-3 Mbps)	Partial alignment (TCP: 5 Mbps, UDP: 6 Mbps sustained)	Strong alignment (TCP: 24+ Mbps, UDP: 6-16 Mbps)
Contention	Unstable coexistence; TCP: 2-5 Mbps, UDP: 2-8 Mbps	Partial stability; TCP: 4-6 Mbps, UDP: 6-12 Mbps	Improved coexistence; TCP: 5-24 Mbps, UDP: 6-19 Mbps
Multi-Hop Penalty	50-55% throughput loss	40-45% throughput loss	31% throughput loss



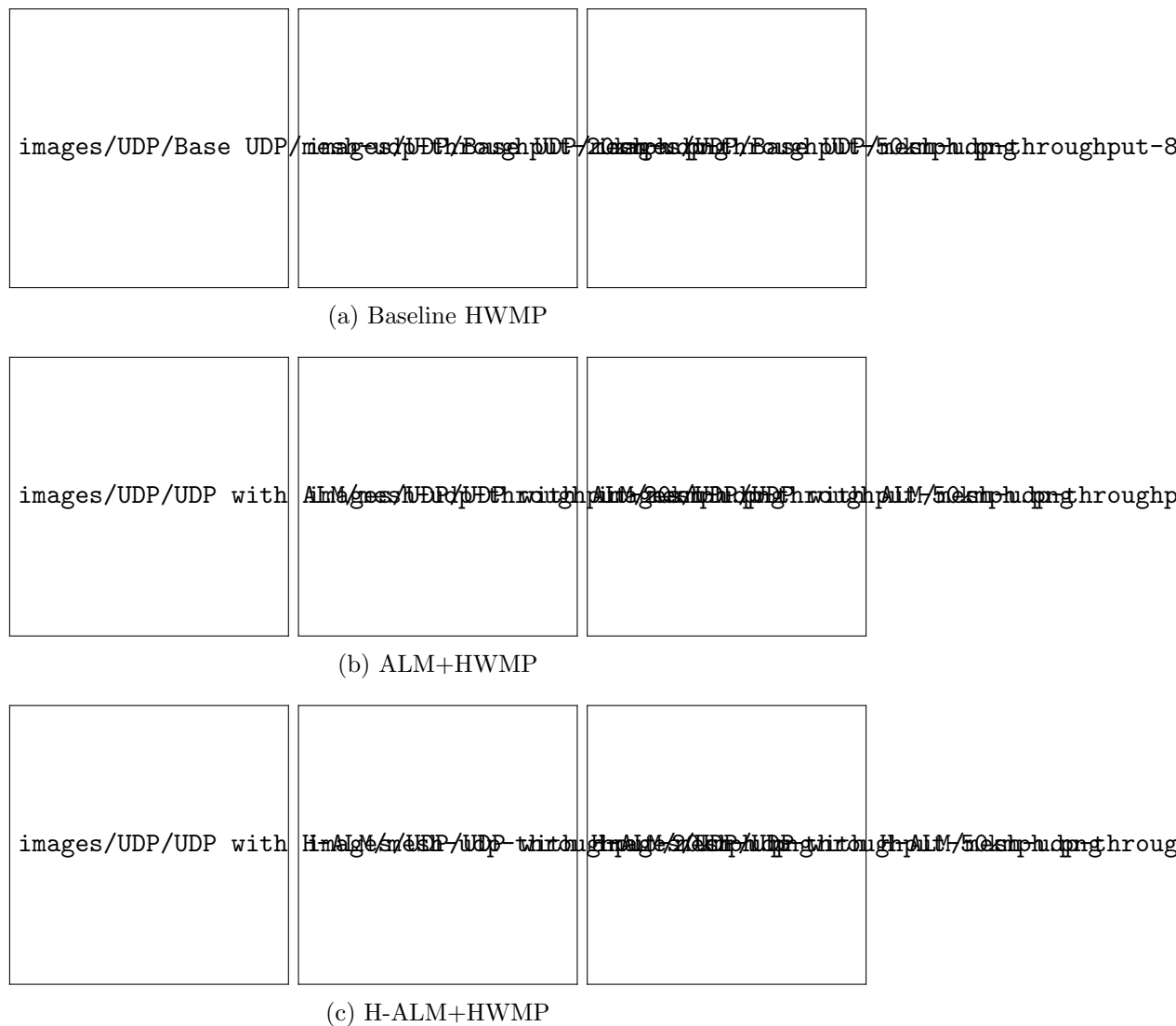


Figure 6: Throughput (UDP)

4.5 Jitter

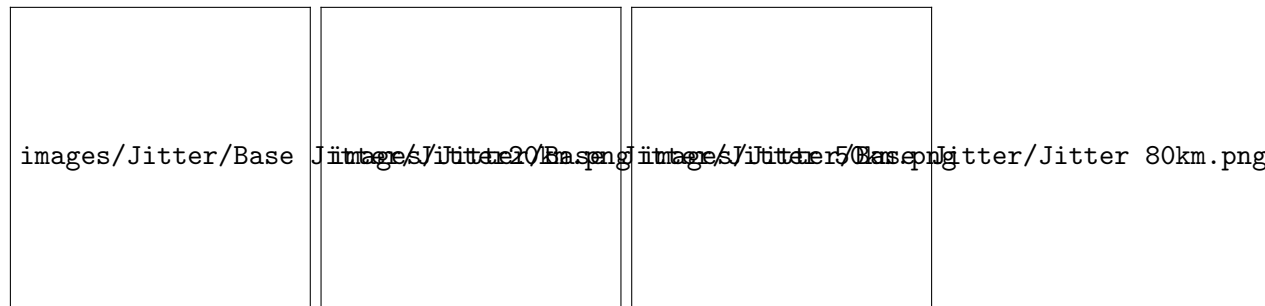
Figure 7 reports jitter for all three routing schemes over the same range of vehicle speeds. The message is consistent with the RTT and throughput results: as mobility increases, unstable routing decisions lead to larger delay variation.

Baseline HWMP shows the highest jitter, particularly once the vehicle enters the overlapping region where repeated route changes occur. At 50 and 80 km/h, the delay variation becomes pronounced and occasionally reaches very large values. ALM+HWMP improves the situation somewhat, but visible bursts remain, which indicates that the route-selection process still reacts too strongly to transient changes.

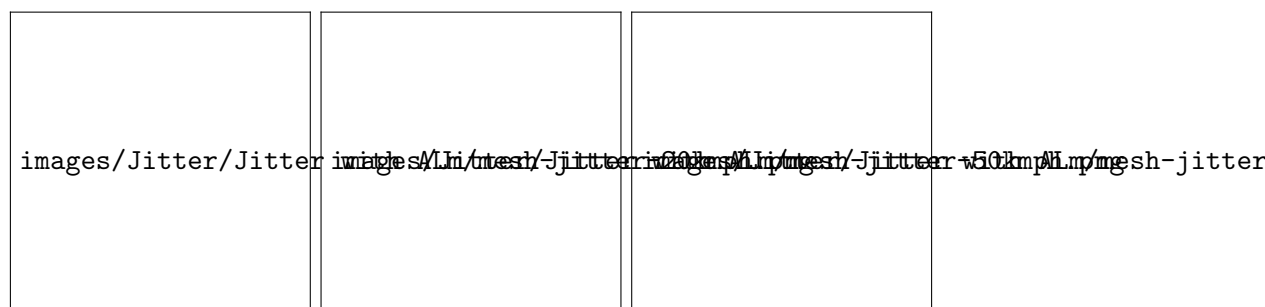
H-ALM+HWMP produces the most controlled jitter traces. Because the hysteresis term suppresses unnecessary switching, delay variation remains within a much narrower band and large excursions are less frequent. This improvement is especially relevant for real-time traffic, where consistency matters as much as average performance.

Taken together with the throughput analysis, the jitter results highlight the cross-layer cost of route oscillation. Repeated handoff does not only change the path; it also destabilizes transport behavior and prolongs recovery. H-ALM reduces those disruptions by roughly 80–85%, allowing TCP and UDP flows to experience a more predictable channel state. Table 3 confirms the same

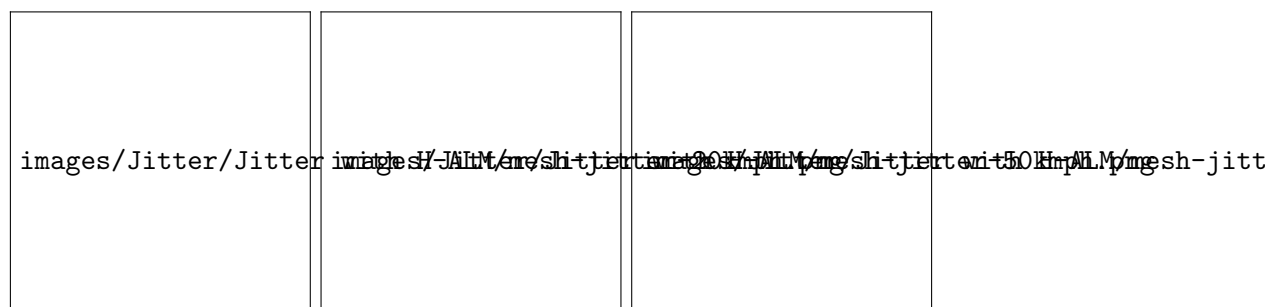
trend at 80 km/h, where the proposed method yields lower peak jitter and smoother handoff behavior. The remaining limitation is that multi-hop forwarding still introduces a throughput penalty, and very short jitter spikes may be underrepresented by the 1 ms iperf sampling interval.



(a) Baseline HWMP



(b) ALM+HWMP



(c) H-ALM+HWMP

Figure 7: Jitter

Table 6: Jitter and Handoff Behavior Comparison across Mobility Protocols at 80 km/h

Parameter	HWMP	ALM + HWMP	H-ALM + HWMP
Peak Jitter (80 km/h)	1200 ms	600 ms	280 ms
Distance Correlation	Strong ($R^2 > 0.8$)	Weak ($R^2 \sim 0.3$)	Very strong ($R^2 > 0.95$)
Handoff Smoothness	Uncontrolled	Partial stability	Substantially improved
Control Overhead	High	Medium	Low
Best Use Case	Static nodes	Moderate mobility	High mobility

5 Conclusion

This research examined hand-off instability in IEEE 802.11s mesh networks using vehicular mobility, specifically looking at "ping-ponging" caused by excessive reliance on the standard air-time metric in response to temporary changes in channels. We used simulations in NS-3 of an edge case where there was a mobile node traveling between two stationary roadside mesh network nodes (as shown above) and concluded that HWMP had a propensity for repeated route switches, significant delay variability, extremely high packet loss rates, and decreased throughputs with increased speeds.

We created H-ALM, a hysteresis based extension of the air-time link metric that uses both the current observation and a previous value once it has been determined as stable. Although the design of H-ALM is relatively simple; it provides a marked improvement in terms of overall system performance. In all of the simulation scenarios tested, H-ALM reduced packet loss averages by approximately 91% compared to HWMP, kept RTTs in a more stable operating range than either HWMP or ALM+HWMP, limited peak jitter to less than 280ms, and preserved more consistent TCP and UDP throughput.

Our primary conclusion from this study is that modest temporal smoothing in the routing-metric layer can provide material improvements to the behavior of IEEE 802.11s in mobile mesh networks. Future studies could include expanding the study to larger topologies, developing methods for adaptively selecting values for α , performing hardware experiments to support claims made within this paper, and designing predictive handoff mechanisms for more complex vehicular deployments.

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